

Bugs found: 3

1. The static inner class `BadKey` overrides `equals()` but fails to override `hashCode()`, violating the contract that equal objects must have equal hash codes.
2. The variable `k` is initialized to `null`, causing a `NullPointerException` when `k.hashCode()` is called.
3. Accessing index `-1` of the array `'a'` will throw an `ArrayIndexOutOfBoundsException` as array indices must be non-negative.